

Module 02 · Lecture 2

How to design a HCI system?

Users, stakeholders, conceptual design, and why we sketch

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Interação Pessoa-Máquina · DCC · FCUP

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Before you even begin to plan



Choose colleagues and form the group

You need 4 people



Everyone should understand their role and what is expected

Roles rotate: team leader in every session, one panel per member



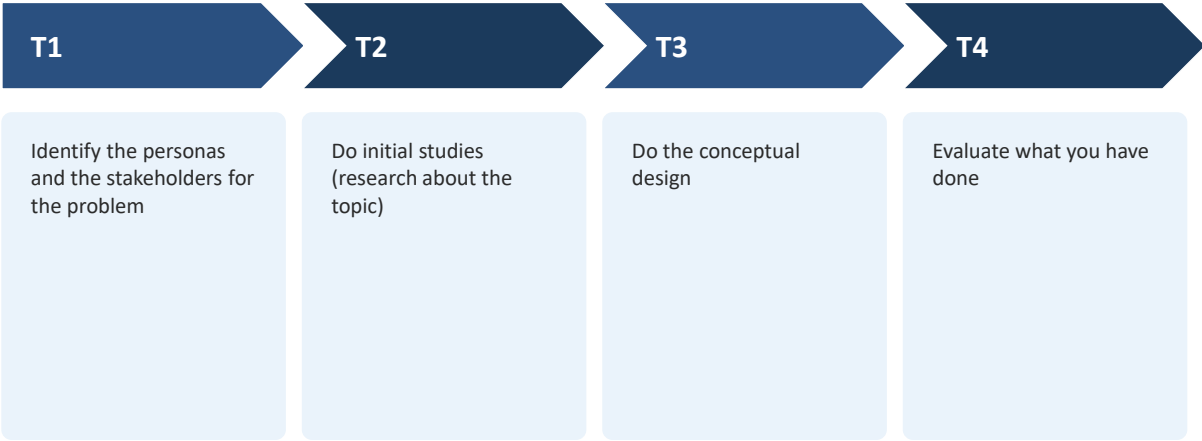
Discuss group management

Meetings, milestones, progress indicators, where the evidence is kept

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The tasks: general idea

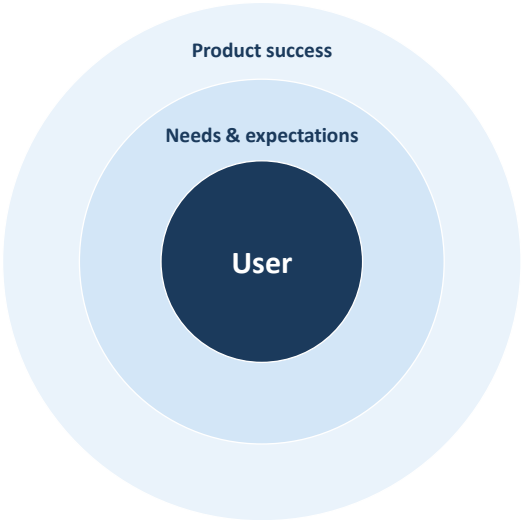


Four tasks, in this order, and then back to T3 with what T4 tells you.

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Identify users

- The users that ultimately will use the product
- These users have needs and expectations (often not consciously)
- Their experience with the product will dictate the actual product's success
- **Therefore, the user should be in the centre of the process: *User-Centred Design***
- *Create personas to objectively describe user possibilities*



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T1

Identify the users and the stakeholders

- “Stakeholders” are everyone who cares about the system
- Think:
 - Who is going to use it? (*personas*)
 - Who is going to build it?
 - Who is going to sell it?
 - Anyone else who might care?
- How?
 - Brainstorm within the group. Discuss it with your friends and colleagues. Write them down on a list



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T1.1

User research

Focus on:

-  potential users
-  their characteristics
-  the tasks they need or want to accomplish
-  the context in which they operate
-  their expectations

Where the answers come from

Interviews and observation (next slides), short questionnaires, existing data, and the stakeholders you listed in T1.

Write down the users' own words: they are the vocabulary of your conceptual model.

Based on slides by Rui Rodrigues (FEUP) and Bruno Giesteira (FBAUP)

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How are they defined?



What they know about the tasks

- Previously known, recently introduced, or new
- How they have understood them
- Their level of expertise



What they know about the tools

- What they currently use
- How they use them



What mental models they have

- How they interpret the system internally
- What vocabulary they associate with elements and tasks → conceptual model

Based on slides by Rui Rodrigues (FEUP) and Bruno Giesteira (FBAUP)

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How do they differ individually?

Within a user population there are individual differences that may lead to sub-groups of users



Personal traits

e.g. learning methods or work methods



Physical differences

Visual acuity, hearing, disabilities, age-related features



Cultural differences

LTR vs RTL languages, expressions, symbols, age-related (e.g. teenager vs adult)



Motivation and attitude

Eager for new things vs. resistance to change

Based on slides by Rui Rodrigues (FEUP) and Bruno Giesteira (FBAUP)

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Their expertise

Beginner

- Doesn't know initially what to do
- Afraid to fail
- May become an advanced beginner in a relatively short period

Advanced beginner

- Loses fear
- Knows enough to execute the needed tasks, and focuses on that
- 80% of the users don't go over this level**
- This is the level for which to optimise the interface design**

Competent

- By longer exposure creates a solid mental model
- Can predict and plan better how to execute a task
- Can diagnose and solve problems
- Recognises that is willing to learn more to do more and better

Expert

- Highly motivated
- Uses the product as an integral part of their activity
- Creates own processes, keeps pushing the system
- Typically a very small fraction of the user base

How to do this research about users?



Asking users: good and bad questions

Avoid

Leading questions

“Don't you think the new menu is easier?”

Yes/no questions that close the conversation

“Do you like it?”

Double-barrelled questions

“Was it fast and easy to find?”

Hypotheticals

“Would you use this feature?”

Your own jargon

“How do you feel about the onboarding flow?”

Prefer

Open questions

“Tell me about the last time you...”

Behaviour, not opinion

“What did you do?” instead of “What would you do?”

One thing at a time

Split the question, ask the first, then ask the second half

Concrete and recent episodes

“Walk me through what happened.”

The user's own words

Write down their vocabulary: it is your conceptual model

Keeping the conversation going

Conversational prompts

“Tell me more.”

“Why?” (and then why again)

“What happened next?”

“Can you show me?”

“What did you expect to happen?”

...silence. Count to five before the next question.

A minimal interview protocol

 A script of 6 to 8 open questions, 30 minutes, semi-structured: follow what the person says

 Two interviewers: one asks, the other takes notes (quotes, surprises, vocabulary)

 Ask permission before recording. Warm up with context questions

 End with “Is there anything I should have asked?”

 Write the summary right after the interview, while it is fresh

Bias in user research: yours and theirs



Confirmation bias

You notice what confirms what you already believe.

Antidote: Write down, before the interview, what would prove you wrong.



Leading (interviewer) bias

The question already contains the answer.

Antidote: Neutral wording, the same script for everyone.



Social desirability

People say what makes them look good.

Antidote: Ask about facts and recent behaviour, not virtues.



Acquiescence

People tend to agree with whatever you propose.

Antidote: Avoid yes/no. Offer real alternatives.



Recall bias

Memory is reconstruction, not replay.

Antidote: Ask about the last time, never about “usually”.



Sampling bias

Your friends and classmates are not your users.

Antidote: Reach the personas you defined, not the people at hand.

Observe more than you ask. When you can, watch people doing the task.

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Personas

“A Persona is a user archetype you can use to help guide decisions about product features, navigation, interactions, and even visual design”

Perfecting Your Personas, Kim Goodwin

“Personas provide us with a precise way of thinking and communicating about how users behave, how they think, what they wish to accomplish, and why.”

Alan Cooper

“Personas help a team avoid designing for themselves.”

Alan Cooper

A Persona is not...

- Based on demographics or market segments
- Drawn from gut feelings about your audience
- User profiles or stereotypes

From Adaptive Path's case study on PayCycle

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10 rules to create personas

1. Keep them simple and **memorable**

2. They **should be separated by goals**, not behaviour

3. Focus on satisfying the widest audience, not the sales audience

4. Add some personal detail

5. Focus on 3 or 4 goals per persona
6. Create personas in the context of a specific project

7. Personas **represent behaviour patterns**, not job or role descriptions

8. Keep your persona set small

9. There is not a direct correlation between market segments and personas

10. **Focus on goals, not tasks.** Tasks are things we do to achieve goals

Examples in: <https://venngage.com/blog/user-persona-examples/>

T2

Initial studies

-  **User**

What does the end-user know? What does he want? What is he capable of doing? What metaphors does he recognise? What is his technological awareness?
-  **Technology**

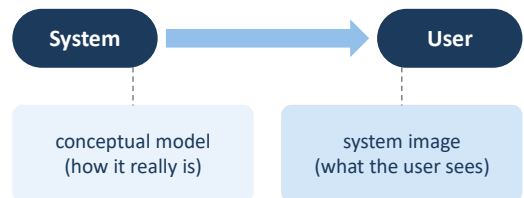
What technology can be used? What characteristics does it have? Which alternatives can be used?
-  **Objectives**

What should the system do? Is something critical? Is something optional? What are the priorities of the various objectives?

T3

Conceptual design

- **Mental models**
- **Design the system functionality**
 - Define how the system really is
- **Build a conceptual model**
 - Define how you want the user to see the system
- **Build a system image**
 - Define an image that will make the user create the intended conceptual model



The user only sees the system image. Sketches and prototypes are the cheapest way to test whether that image produces the right conceptual model.

Tip: use a rapid and simple prototype

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Design is an iterative negotiation process

- **There are conflicting objectives in a design task!**
- Stakeholders:
 - **Simple approach:** consider the user as a single stakeholder
 - **Correct approach:** consider all the stakeholders
- System design / engineering / assembling:
 - **Functional objectives must be met (this is the baseline)**
 - User limitations must be taken into account
 - Technology limitations must be taken into account

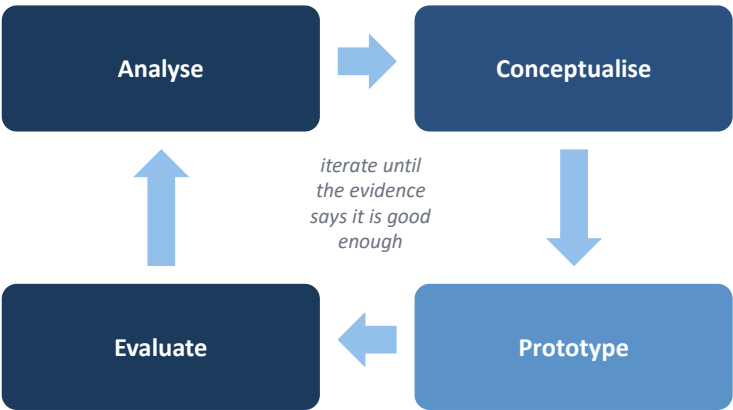


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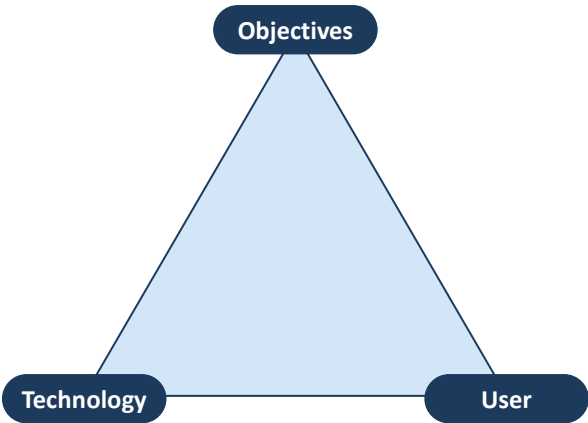
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The iterative design cycle



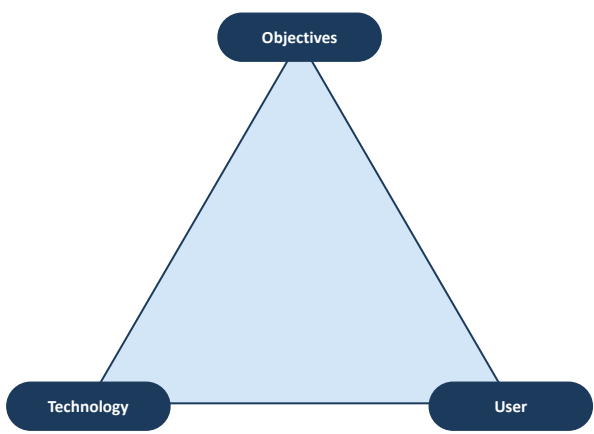
Analyse users, tasks and context → conceptualise (mental model, system image) → prototype → evaluate → back to analysis with what you learned

The magic triangle



Three corners that pull in different directions. A design is the compromise the group negotiates between them.

Tips for your design



- Each of the three (four) group participants studies one corner of the problem
- **In group meetings:**
 - Each member “defends” one “corner”
 - The group “negotiates” a design solution
 - **The negotiation stops when all participants are satisfied with the compromise**

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Sketching is about design

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Sketching is about design

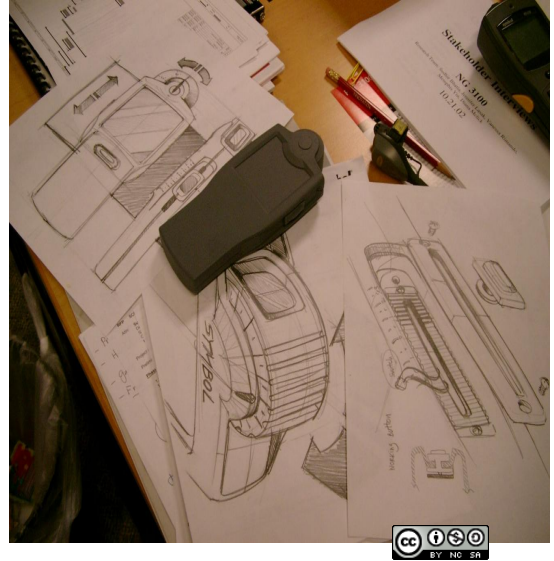
- **Sketching is not about drawing**

- Sketching is a tool to help you:

- express
- develop, and
- communicate design ideas

- Sketching is part of a process:

- idea generation
- design elaboration
- design choices
- engineering



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Why to sketch?



Create

- early ideation
- think openly about ideas
- force you to visualise how things come together
- brainstorming: generate abundant ideas without worrying about quality
- invent and explore concepts



Record

- ideas you develop
- ideas that you come across
- archive ideas for later reflection



Reflect, share, critique, decide

- communicate ideas to others
- invite responses, criticisms, and alternatives
- choose ideas worth pursuing

Very important: avoid mixing ideas on the same sketch

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EXPLORING AN IDEA

Getting the Design Right

Generate an idea

idea 1

idea

Iterate and develop it

idea

idea

idea

idea 1

idea

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EXPLORING AN IDEA

Getting the Design Right

Generate an idea

idea 1

idea

Iterate and develop it

idea

idea

idea

idea 1

idea

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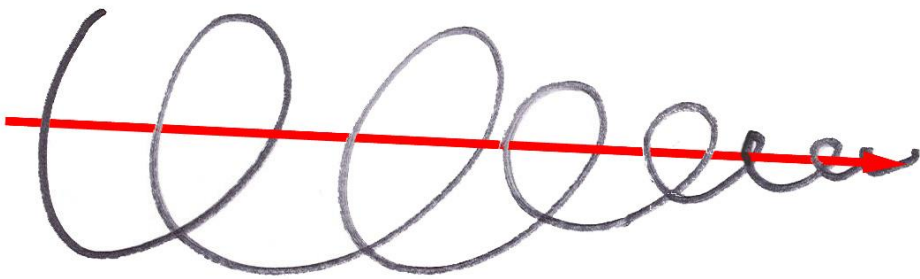
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But is it the best idea?

Getting the Design Right

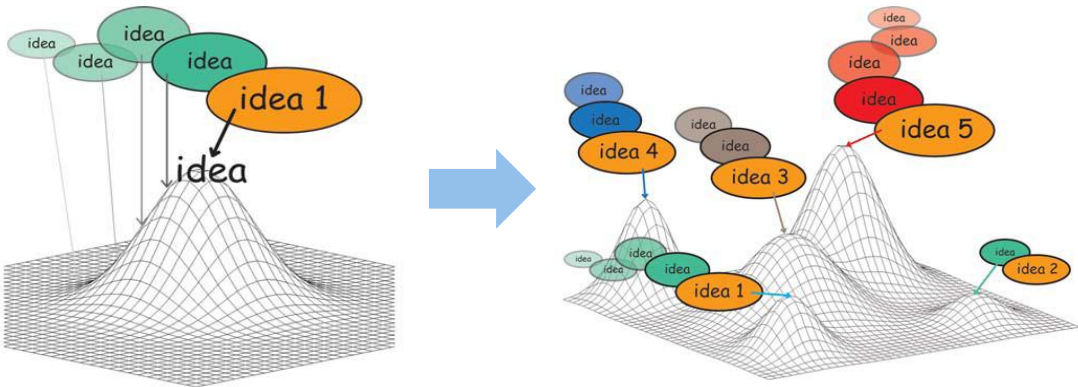


(iterative process)

Slide contributed by Bill Buxton

Getting the Design Right (may not be enough)

The problem: fixates on the first idea. Local hill climbing: did you reach a local or a global maximum?



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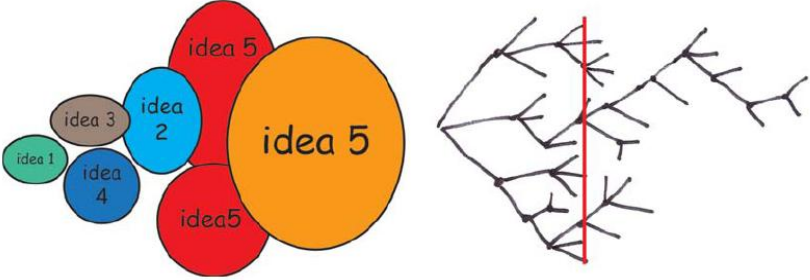


Example: cell phone design



EXPLORING ALTERNATIVES

Getting the Right Design



“... a designer that pitched three ideas would probably be fired. I'd say five is an entry point for an early formal review (distilled from 100's) ... if you are pushing one, you will be found out, and also fired ... it is about open mindedness, humility, discovery, and learning. If you aren't authentically dedicated to that approach, you are just doing it wrong!”

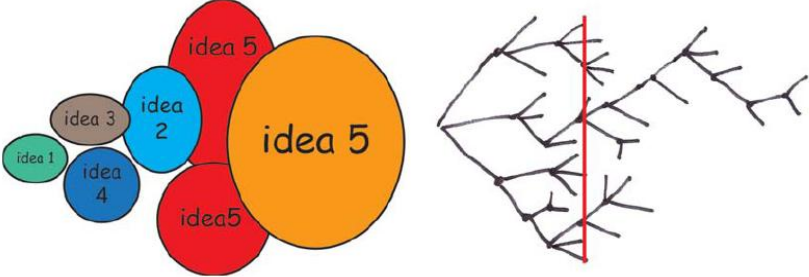
Alistair Hamilton
VP Design, Symbol Technologies

Bill Buxton coined the expression “Getting the Design Right vs. Getting the Right Design”



EXPLORING ALTERNATIVES

Getting the Right Design



- generate many ideas and variations
- reflect on all ideas
- choose the ones that look most promising
- develop them in parallel
- add new ideas as they come up
- then iterate your final choice

Roughly the
"10x10 model"

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Source: Laseau, P. (1980) Graphic Thinking for Architects & Designers. John Wiley and Sons



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Elaboration and reduction

Elaborate

generate solutions. These are the opportunities

Reduce

decide on the ones worth pursuing

Repeat

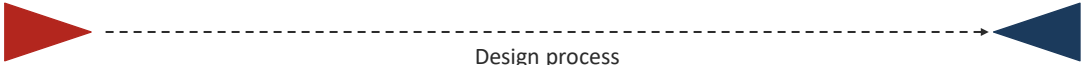
elaborate and reduce again on those solutions

Elaboration

opportunity seeking

Reduction

decision-making



Design process

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Source: Laseau, P. (1980) Graphic Thinking for Architects & Designers. John Wiley and Sons



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Elaboration and reduction

Design is *choice*.

There are two places where there is room for creativity:

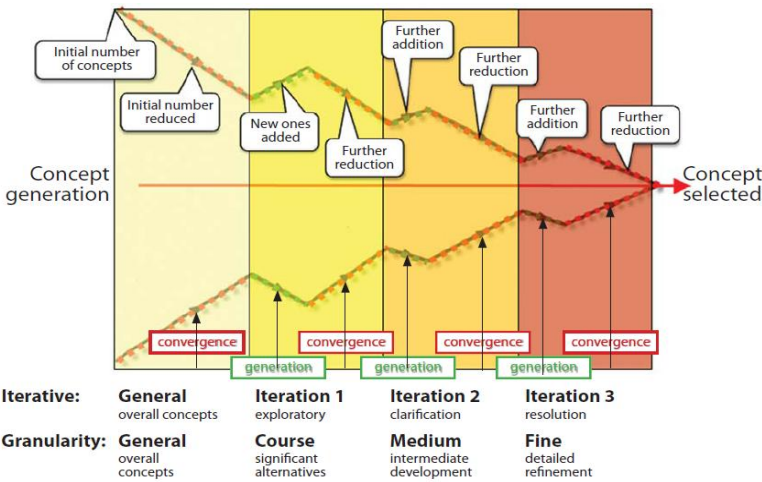
- 1

the creativity you bring to **enumerating meaningfully distinct** options from which to choose
- 2

the creativity you bring to **defining the criteria, or heuristics**, according to which you make your choices

The design funnel

Alternate generation of ideas and convergence until resolution



TPC (homework)

What is the difference between

getting the design right

and

getting the right design?

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T4

Evaluation

- How will you know that your conceptual model is good enough?
- How will you know that your system's image is good?

- **Evaluate!**

- Prototype
- Use adequate evaluation measures and procedures
- Measure
- Assess



in this course:
cognitive walkthrough →
heuristic evaluation →
usability tests

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Final tips



Plan ahead!



Whenever possible, use solid, well-studied methodologies

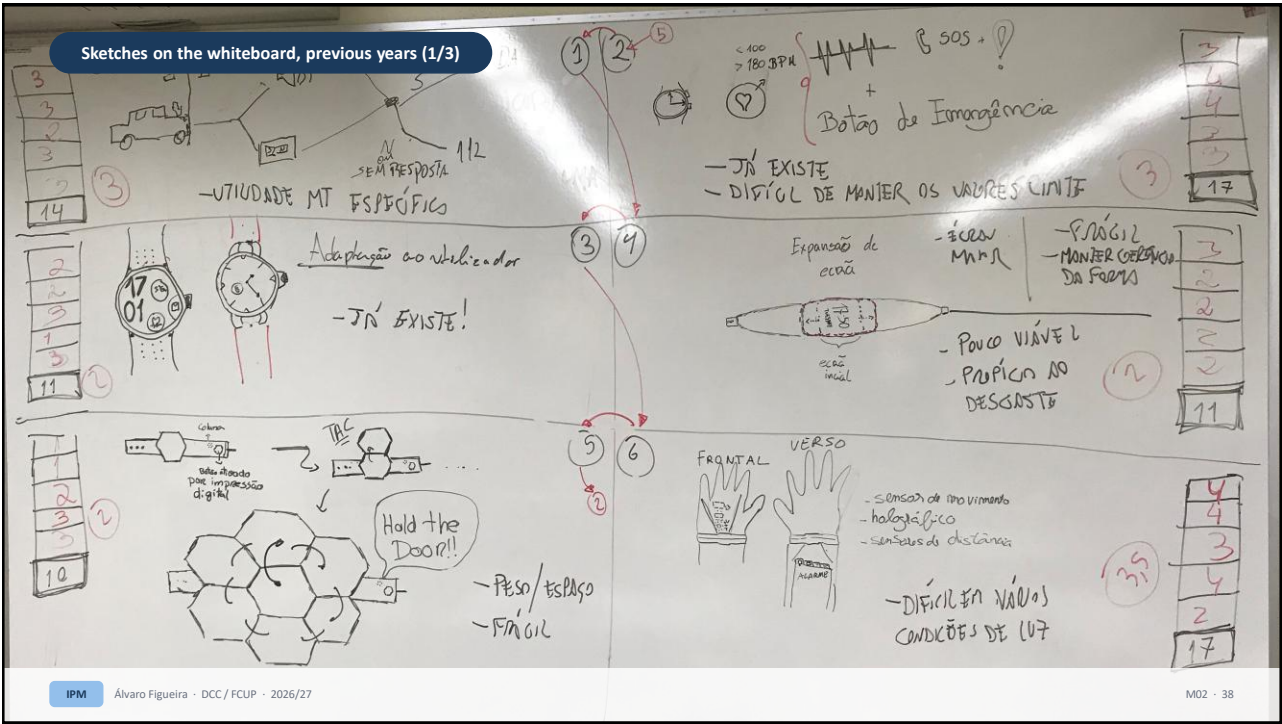
Conceptual design, contextual inquiries, horizontal prototypes, semi-structured interviews, discount evaluation methods, etc.

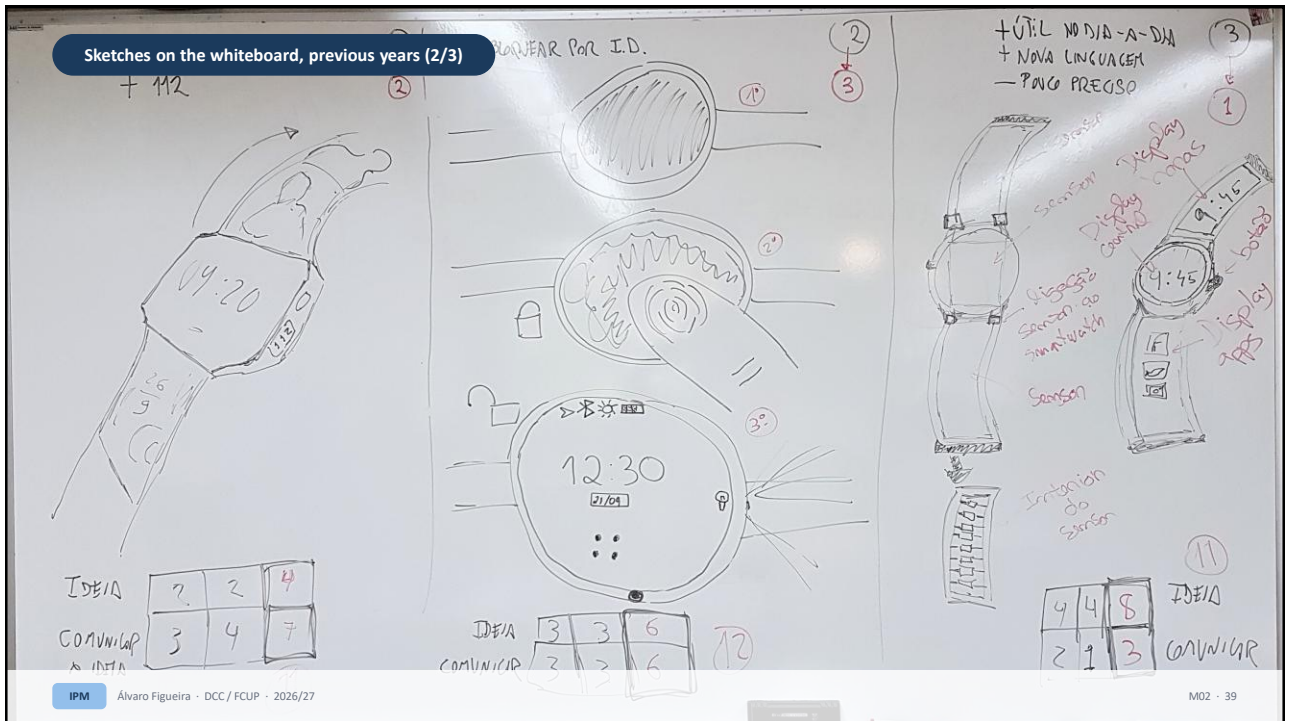


Pay attention in class and use the slides for reference

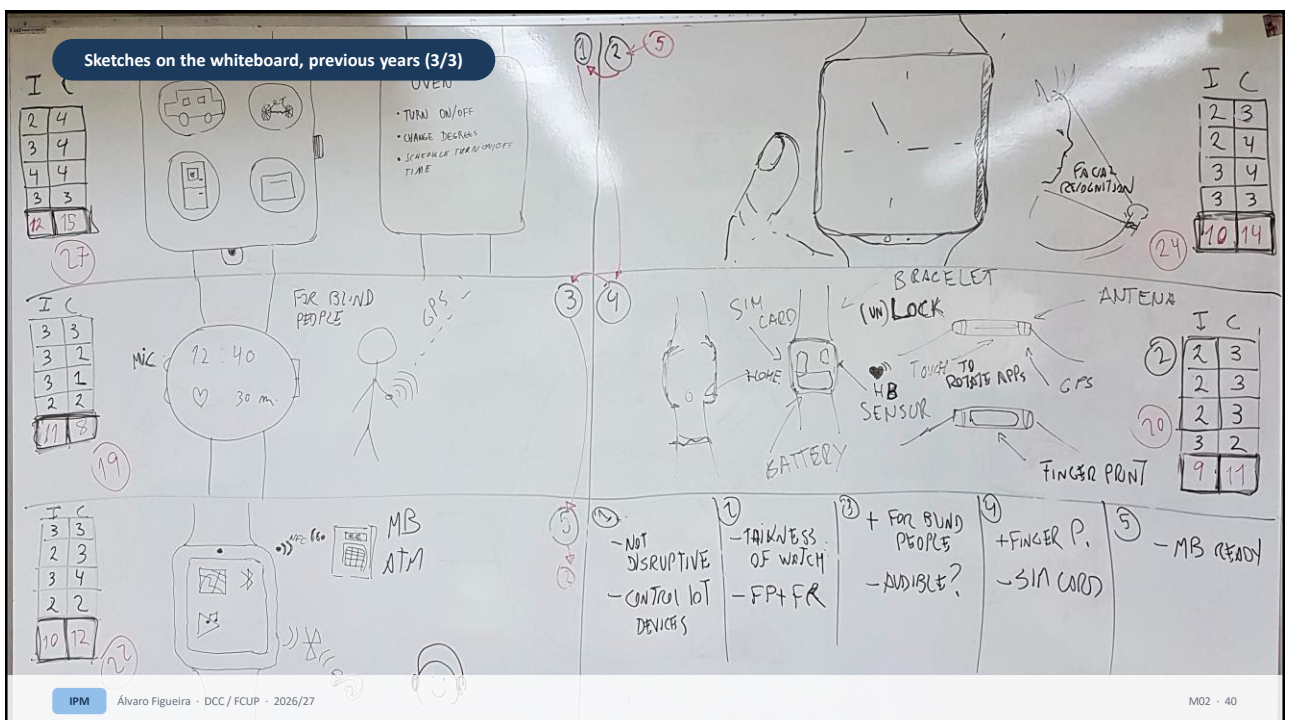


Use the lab classes to discuss your project with the teacher





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